

21-701: Discrete Mathematics

Lecturer: Wes Pegden

Scribe: Sidharth Hariharan

Carnegie Mellon University - Fall 2026

This version of the notes was last compiled on August 24, 2026.

For the latest version, visit [https:](https://thefundamentaltheor3m.github.io/DiscreteMathNotes/main.pdf)

[//thefundamentaltheor3m.github.io/DiscreteMathNotes/main.pdf](https://thefundamentaltheor3m.github.io/DiscreteMathNotes/main.pdf).

For comments, errors or suggestions, please raise an issue/make a pull request at

<https://github.com/thefundamentaltheor3m/DiscreteMathNotes>.

Contents

Course Introduction and Overview	2
1 Graphs and Colourings	3
1.1 Colourings	3
1.1.1 2-Colourings of Graphs	3
1.1.2 Colourings of Hypergraphs	5
2 Another Chapter	7
2.1 Introducing the Main Object of Study in this Chapter	7
2.2 Another Section	7
Bibliography	10

Course Introduction and Overview

This course will cover the theory of absolutely everything mathematical.

Chapter 1

Graphs and Colourings

Much of this course will be about graphs, and a surprising amount of what we want to know about a graph can be phrased as a question about colouring it. We begin with the vocabulary, and then settle the first such question completely: which graphs admit a colouring with only two colours. We then ask the same question of hypergraphs, where it becomes considerably harder—even pinning down how many edges a 3-uniform hypergraph needs before two colours are no longer enough takes real work.

Notation.

- $[k]$ will denote $\{1, \dots, k\}$.
- $\sim$ will denote the adjacency relation on graph vertices.

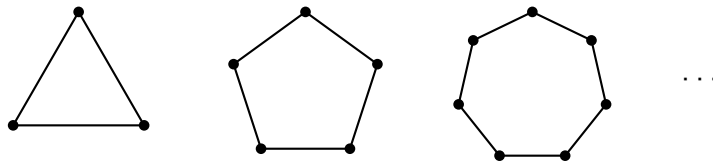
1.1 Colourings

Definition 1.1.1 (Colouring). Given a graph $G = (V, E)$, a k -colouring of G is a map $c : V \rightarrow [k]$ such that $u \sim v \implies c(u) \neq c(v)$.

1.1.1 2-Colourings of Graphs

Let's start with a motivating question: which graphs can be 2-coloured?

Any odd cycle graph can't be 2-coloured. That is, any graph that looks like



Let's study when odd cycles exist.

Lemma 1.1.2. *If G has an odd closed walk, then G has an odd cycle.*

Recall that a closed walk need not be a cycle: walks can contain repetitions. So even line graphs can have closed walks (start from one vertex, go to its neighbour, return).

Proof. Suppose G has an odd closed walk. Let W be such a walk of minimal length. Write $W = (x_1, \dots, x_i, \dots, x_j, \dots, x_\ell)$, with $x_\ell = x_1$. Suppose W is not a cycle. Then there must be some x_i, x_j such that $x_i = x_j$ even though $i \neq j$, ie, there must be some repetition. Then, $(x_1, \dots, x_i, x_{j+1}, \dots, x_\ell)$ and $(x_i, x_{i+1}, \dots, x_j)$ are both shorter walks than W . Moreover, one of them must be of odd length. This contradicts minimality of the length of W . So W must be a cycle. □

Here's a cool result.

Theorem 1.1.3. *A graph $G = (V, E)$ can be 2-coloured if and only if G has no odd cycle.*

Proof.

($\implies$) We've already seen that odd cycles can't be 2-coloured. So if G can be 2-coloured, it can't have an odd cycle. We've essentially seen this from the examples above.

($\impliedby$) We may assume G is connected, since otherwise we can colour each component separately.

Fix some $v_0 \in V$. Define $c : V \rightarrow \{1, 2\}$ by

$$c(u) = \begin{cases} 2 & \text{if } \text{dist}(v_0, u) \text{ is odd} \\ 1 & \text{if } \text{dist}(v_0, u) \text{ is even} \end{cases}$$

where $\text{dist}(\cdot, \cdot)$ denotes the shortest number of edges from one vertex to another.

We claim that this c is a valid 2-colouring.

Indeed, suppose it is not. Then, there must be some $u, v \in V$ such that $u \sim v$ and $c(u) = c(v)$. Now let's consider the two possible cases on the colour.

Suppose $c(u) = c(v) = 1$ for adjacent u, v . Then, there is a path of even length between v and v_0 and another path of even length between u and v_0 . Concatenating the path from v to v_0 with the one from v_0 to u , and closing up with the edge uv , we get a closed walk of length $\text{even} + \text{even} + 1$. This is odd, so by Lemma 1.1.2, G has an odd cycle.

The case $c(u) = c(v) = 2$ is identical: the two paths are now of odd length, so the closed walk they form with the edge uv has length $\text{odd} + \text{odd} + 1$, which is odd again.

Either way, G has an odd cycle, contrary to hypothesis. So c must be a valid 2-colouring.

□

1.1.2 Colourings of Hypergraphs

Colouring is not really a question about graphs. All the definition above asks of G is that we know which pairs of vertices may not share a colour, and nothing stops us forbidding larger sets instead.

Definition 1.1.4 (Hypergraph). Given a vertex set V , a **hypergraph** is a collection H of subsets of V . The sets in H are called **hyperedges**.

Definition 1.1.5 (Uniformity). A hypergraph H is **k -uniform** if $|f| = k$ for all $f \in H$.

The following is thus obvious.

Proposition 1.1.6. *A graph is a 2-uniform hypergraph.*

Definition 1.1.7. A proper colouring of a hypergraph H is an assignment $c : V \rightarrow [r]$ such that $\forall f \in H, |c(f)| \geq 2$.

We say “no edge is monochromatic”.

We can ask ourselves the following question: **“Which 3-uniform hypergraphs are 2-colourable?”**

For each k , define $m(k)$ to be the smallest m such that $\exists$ some k -uniform hypergraph with m edges that is not 2-colourable. We can show that $m(2) = 3$. But what is $m(3)$? That is, how many 3-edges do I need to prevent 2-colourability?

Let H be a 3-uniform hypergraph that’s not 2-colourable. Assume, further, that H has at most 6 edges (each being a subset of size 3).

First, we’ll show that we can reduce, without loss of generality, to the case where H has exactly 6 vertices. Indeed, if H has vertices x, y that are not in any common edge, then I can define H' by “merging” x and y . So H' has one fewer vertex than H . This H' cannot be 2-colourable: if it were, then that colouring would extend to H by using the colour of the merged vertex at both x and y , which would make H 2-colourable. And when H has more than 6 vertices, such a pair always exists.

To see this, call a pair of vertices **covered** if some edge of H contains both of them; an uncovered pair is exactly what we are after. Now count. An edge has 3 vertices, so it covers the $\binom{3}{2} = 3$ pairs among them, and H has at most 6 edges, so at most

$$6 \cdot \binom{3}{2} = 18$$

pairs are covered in total. But if H has $n \geq 7$ vertices, there are $\binom{n}{2} \geq \binom{7}{2} = 21$ pairs to cover. So some pair is left uncovered, and merging it brings us one vertex closer to 6.

So now assume that H has exactly 6 edges and 6 vertices and is not 2-colourable.

Chapter 2

Another Chapter

You get the idea.

2.1 Introducing the Main Object of Study in this Chapter

Woah. Very cool.

2.2 Another Section

Yup, \lipsum time. Boy do I love L^AT_EX!

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut

lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Bibliography

These lecture notes are based heavily on the following references:

- [1] Nicolas Bourbaki. *Groupes et Algèbres de Lie. Chapitre 1*. 1st ed. Éléments de Mathématique. Édition originale publiée par Hermann, Paris, 1972. Springer Berlin, Heidelberg, 2007. ISBN: 978-3-540-35337-9. DOI: <https://doi.org/10.1007/978-3-540-35337-9>.

Visit <https://thefundamentaltheorem.github.io/DiscreteMathNotes/main.pdf> for the latest version of these notes. If you have any suggestions or corrections, please feel free to fork and make a pull request to the [associated GitHub repository](#).